Datasets

We will mainly use GLAMOS datasets (from the Swiss glacier monitoring service), combined with a swisstopo base map.

The Swiss Glacier Inventories (1850, 1931, 1973, 2010, 2016) are provided as ESRI shapefiles and contain glacier geometries (polygons), along with metadata such as area, elevation, slope. These inventories allow reconstruction of glacier extent over time for all Swiss glaciers. Interestingly, the number of glaciers decreased from 3051 in 1973 to 1400 in 2016.

The Swiss Glacier Length Change dataset contains 10’861 observations across 156 glaciers from 1847 to 2024. It provides direct measurement of glacier retreat in meters, with a mean retreat rate of -11.79m per year. This dataset however contains many irregular observations in time, and also includes quality flags. This means that in order to use this dataset, it will need quite a bit of pre-processing (filtering, normalization, ….).

The Swiss Glacier Flow Velocity dataset provides vector measurements of surface ice flow (velocity vectors indicating direction and magnitude), but only for a limited subset of glaciers.

The main derived variable will be a rate-of-change metric, computed either as area change per year (from inventories) or length of change per year (from GLAMOS measurements).

Some preprocessing challenges we foresee are the:

- Aligning of glacier identifiers across inventories
- Handling missing parent links
- Aggregating of irregular time-series observations (older datapoints are inconsistent)

Together, these datasets provide both nationwide coverage and detailed temporal measurements, making them sufficient for the proposed visualisation below.

Problematic

This project aims to visualize glacier retreat across Switzerland, focusing on differences in the rate of change, rather than static snapshots. The goal is to show in a visually intuitive way the magnitude of retreat, and thus raise awareness to the acceleration of glacier retreat.

The target audience we aim to reach is the general population, but also the regulators to push both to better study the effects of climate change.

Existing glacier visualisations typically show glacier outlines at different points in time, or provide per-glacier time series. These do not allow for a direct comparison of retreat rates across all glaciers at a national scale.

We plan to address this by constructing a Switzerland-wide map where each glacier is colored by a normalized rate-of-change metric (% area loss per year).

The visualization will be structured in two levels:

1) Switzerland-wide overview: all glaciers are displayed on a map, colored by the rate of change
2) Focused analysis on the Aletsch Glacier, Matterhorngletscher and the Rhône Glacier: these glaciers have additional datapoints and as such we plan to overlay surface melt flow direction and magnitude to show the local evolution of the glaciers (show vector fields in combination with time series, and/or depending on complexity, animations to show the evolution over time)

Additionally, if time permits, we will explore correlations of glacier retreat with a variety of other additional datasets such as tourism intensity, the average temperature in Switzerland, light pollution maps, …

Exploratory Dataset Analysis

Glacier inventory consistency

The inventory contains between 1400 and 3051 glaciers, depending on the year. Overall data completeness is high (~5% missing values), however missing parent links are quite frequent for older measurements. For example in 1973 2.7% of entries have missing parent links. This will prove a challenge, as to track individual glaciers across time will require careful consideration, especially since glaciers can split and merge.

Glacier size distribution

Glacier are highly heterogeneous in size, for instance in the 2016 measurements the mean area was 0.69km², median area of 0.09km² and the max area 78.49km²

This strong skew implies absolute change is not comparable across glaciers, motivating the use of normalized metrics. We thus plan to use the % area change.

Temporal coverage and data quality

Before 1950, there is almost no reliable data. Between 1950-1990 the data is sparse and inconsistent. After 2000 there is high coverage and data quality.
Thus, our analysis will focus on the modern period, as this is the most reliable period.

Mean length change over time

In the annex you will find two plots we obtained from the data that show the average glacier length over time, and the availability/quality of the data. This illustrates our previous observations about the coverage and quality.

Feasibility

Initial exploration confirms both feasibility and scope limitations of the proposed visualization: this visualization is indeed technically feasible. We do however expect to spend a substantial amount of time on data pre-processing, but we do believe that the end goal warrants this extra effort.

Related Works

The GLAMOS center has published an interactive glacier map viewer with glacier locations, time series and outline comparisons. However, it focuses on per-glacier exploration and does not focus on retreat rates across glaciers at a national level. Our approach addresses this by directly displaying this rate-of-change on the map.

Swisstopo "time-travel" image bank, enables comparison of historical aerial images. While this is a visually compelling way of showing glacier retreat, it provides a qualitative rather than quantitative insight. We plan, if time permits, to integrate this as an optional qualitative add-on when a user clicks on a glacier to complement the visualization.

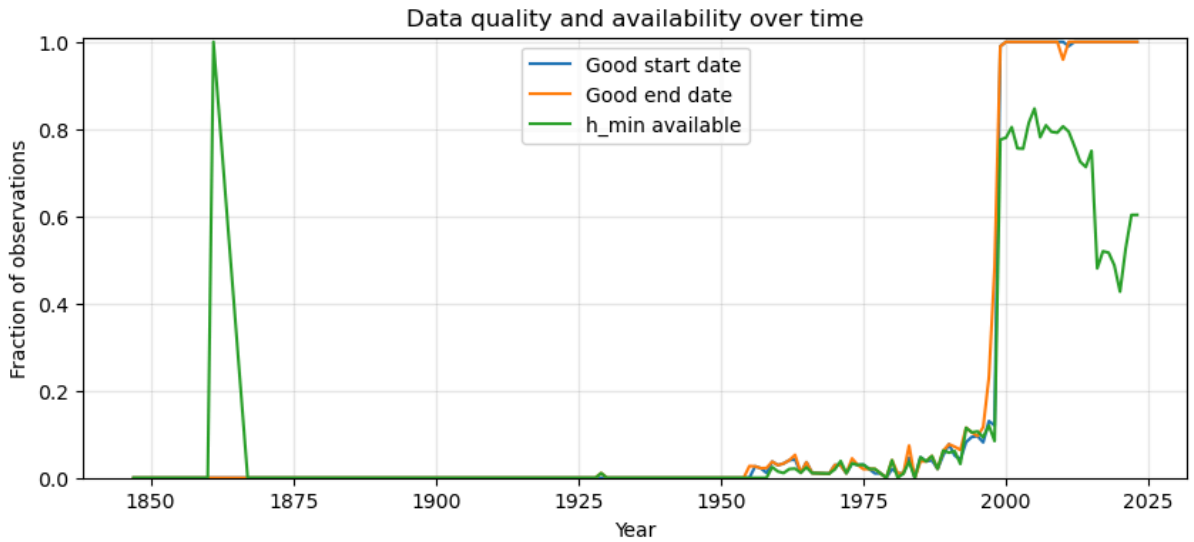
A few storytelling blog style projects exist, such as this one on the Rhone Glacier, but they tend to focus on individual glaciers, and do not use the flow rate in their visualizations. While they are an effective narrative tool, they lack the global glacier comparison and do not encode retreat rates directly into the visualization.

Our approach differs in that we combine:

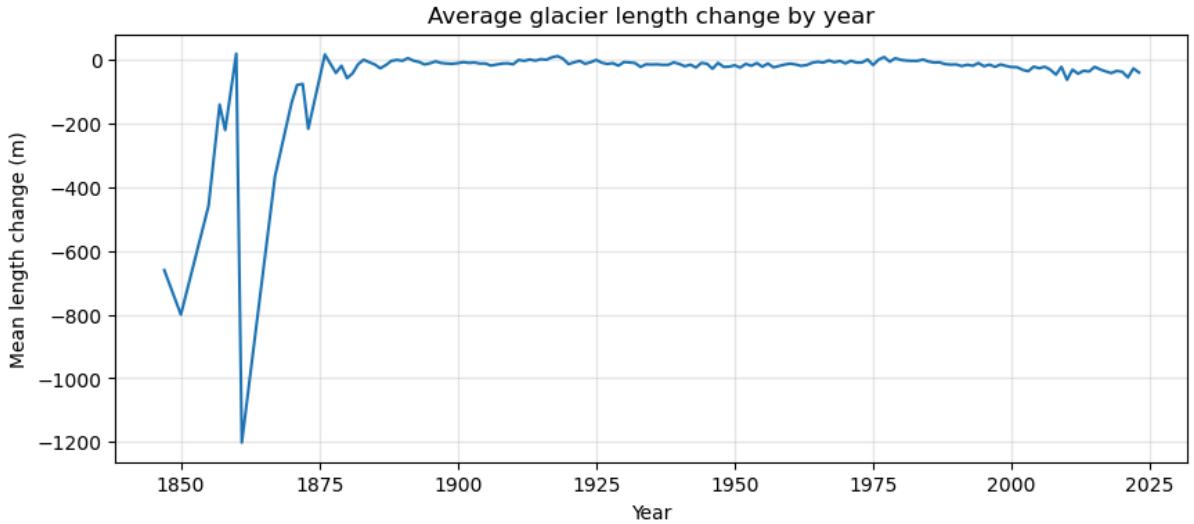
- a nationwide comparative view of all glaciers
- a quantitative, visual, encoding of retreat rates directly into the map
- a local dynamic (possibly even animated) indicator of flow direction and magnitude for selected glaciers

This unique combination strives to show visually the acceleration of the retreat rates, which is a worrying phenomenon. The goal is to emphasize not only that glaciers are retreating, but also the magnitude of how quickly this change is occurring.

Annex

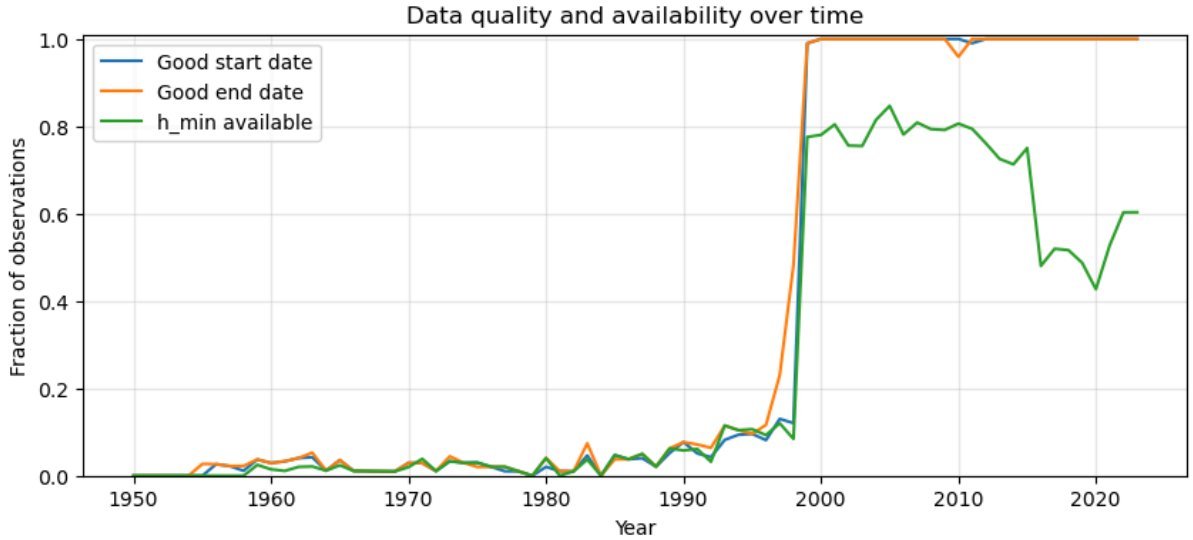


This plot shows the fraction of observations over time for three indicators: valid start date, valid end date, and availability of minimum elevation (h_min); **best data after 2000**.

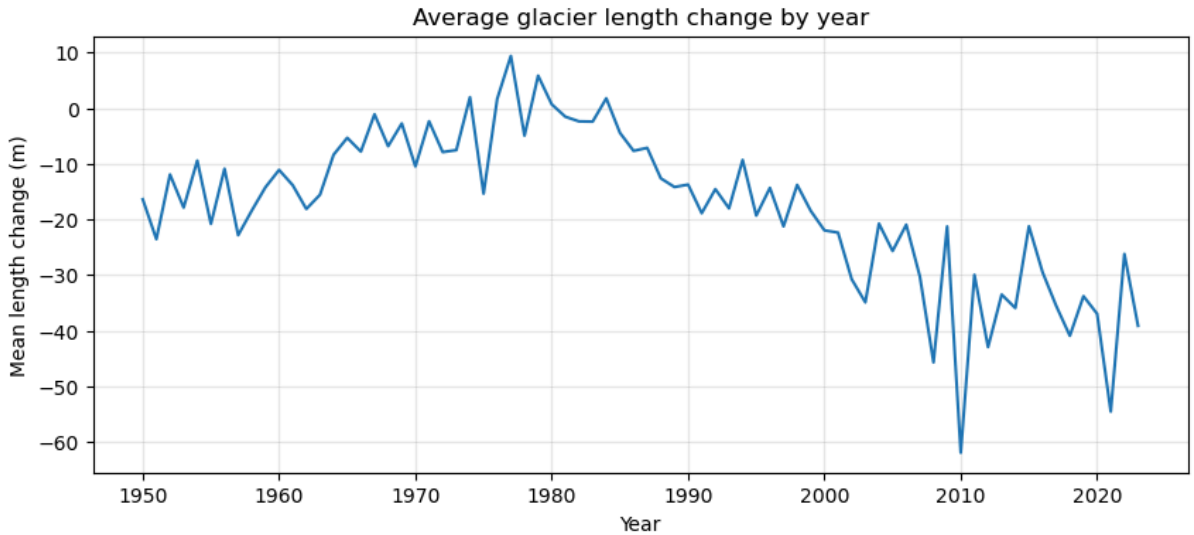


This plot shows the average glacier length change per year (in meters) over time, computed across all available observations.

And the following two graphs are the result of truncating the data before 1950.



This plot shows the fraction of observations over time for three indicators: valid start date, valid end date, and availability of minimum elevation (h_min); **best data after 2000**.



This plot shows the average glacier length change per year (in meters) over time, computed for observations after 1950.